Mahimul Islam

Full Stack Software Engineer & AI Researcher

MS in Computer Science, Georgia Institute of Technology

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Links: LinkedIn | Portfolio | GitHub

Summary

Full Stack Software Engineer with 6+ years of experience building and scaling production systems across healthcare, manufacturing, e-commerce, and B2B SaaS. Experienced owning the full software lifecycle from system architecture and API development to cloud deployment, internal operations, and customer-facing products using Python, TypeScript, React, Next.js, Node.js, PostgreSQL, AWS, and Odoo. Applied AI researcher with work in metacognitive AI agents, knowledge-based AI, LLMs, and Task–Method–Knowledge architectures through Georgia Tech's Design Intelligence Lab.

Skills

Languages: Python, TypeScript, JavaScript, SQL

Frameworks: React.js, Next.js, Node.js, Flask, Django, Vue.js, Odoo OWL

Databases: PostgreSQL, Neo4j

Cloud and DevOps: AWS (ECS, Fargate, EC2, S3, RDS), Docker, CI/CD (GitLab CI/CD, Bitbucket Pipelines)

AI/ML: OpenAI APIs, LLMs, Knowledge-Based AI, TMK architectures, AI agents

Other: REST APIs, JSON-RPC, SSE, SendGrid, Excalidraw, Jira, Confluence, Selenium, Cypress, Jest, Mocha

Experience

Full Stack Software Engineer & AI Researcher, MAEKNIT, Inc – Jamaica, NY — Mar 2025 – Present

- Lead engineering execution across a team of four engineers, owning core architecture, implementation, deployment, and production support for MAEKNIT's client, factory, and internal platforms while mentoring team members and working directly with the CEO and CTO.
- Designed and led development of a Multi-Portal RBAC Architecture for Admin, Client, and Factory users using Next.js, React.js, and TypeScript, supporting collections, garments, samples, swatches, documents, quotations, and collaboration workflows.
- Integrated MAEKNIT's software platform with Odoo 18 Enterprise, developing custom Python and OWL functionality and JSON-RPC integrations across CRM, Sales, R&D, MRP, Inventory, and warehouse operations.
- Built a Full CRM and Client Onboarding Pipeline with bidirectional Odoo synchronization, automated portal account creation, credential delivery through SendGrid, activity tracking, and collection assignment.
- Built AI-assisted client workflows using OpenAI APIs to summarize meeting transcripts, structure project requirements, and generate draft client proposals surfaced through the internal and client portals.
- Developing KnitDraft / TMK-Knit, a human-in-the-loop AI research system applying Task–Method–Knowledge modeling to assist industrial flat-knitting machine program generation while keeping expert review and validation in the workflow.
- Designed the KnitDraft data pipeline around machine programs, panel specifications, measurement data, swatch results, program versions, timing information, and manufacturing knowledge captured through MAEKNIT's development workflow.
- Developed a Document and Asset Management System using AWS S3 presigned uploads, client visibility controls, reusable assets, project boards, and Excalidraw-based visual collaboration.
- Built real-time collaboration features using Server-Sent Events (SSE), including project messaging, activity updates, and cross-portal synchronization.
- Engineered a Factory Quotation System allowing internal teams to broadcast requirements to manufacturing partners, collect competitive quotations, and connect accepted work into downstream Odoo Sales, R&D, and MRP workflows.

- Technologies: Python, Odoo, OWL, JavaScript/TypeScript, Next.js, React.js, Node.js, AWS (ECS, Fargate, RDS, S3), PostgreSQL, Docker, SendGrid, Excalidraw, SSE, JSON-RPC, OpenAI APIs.

Full Stack Engineer, Genome Medical Inc – South San Francisco, CA Aug 2022 – Jan 2025

- Built and maintained healthcare applications across Admin, Patient, and Provider portals using React.js, Vue.js, Python Flask, Node.js, and PostgreSQL in a HIPAA-regulated environment.
- Enabled secure patient data migration between Genome Medical and GeneMatters with zero data loss using a React.js + Redux workflow and secure REST APIs.
- Architected and deployed cloud-based services using AWS ECS, Fargate, EC2, S3, RDS, Docker, Flask, Node.js, and PostgreSQL.
- Strengthened platform security by identifying and resolving more than 50 vulnerabilities using Aikido and Sentry.
- Designed and maintained CI/CD pipelines using GitLab CI/CD and Bitbucket Pipelines, automating build, testing, and deployment workflows.
- Designed role-based access control systems with separate permission tiers for administrators, clinical staff, patients, providers, and partner users.
- Built automated test suites using Mocha, Jest, Selenium, and Cypress, improving release reliability and reducing production defects.
- Provided production support across customer-facing and internal systems, resolving more than 150 support tickets within six months.
- Worked in Agile product teams through sprint planning, refinement, retrospectives, Jira tracking, and Confluence documentation.
- Technologies: React.js, Vue.js, TypeScript, JavaScript, Python Flask, Node.js, AWS, Docker, PostgreSQL, Redux, REST APIs, SSE, SendGrid, Aikido, Sentry, GitLab CI/CD, Bitbucket Pipelines, Jira, Confluence.

Software Developer, Cardinal Selling Services – Huntingburg, IN Jul 2014 – Jan 2021

- Developed websites, internal tools, and REST-based web services using Python, Django, JavaScript, HTML, CSS, and Bootstrap.
- Conducted user requirement analysis and designed relational database structures and Entity Relationship Diagrams.
- Built and maintained data scraping pipelines and automation tools using Python.
- Conducted automated testing using Mocha, Jest, Selenium, and Cypress.
- Worked with relational database design, SQL, and PostgreSQL.
- Used Linux, Git, pull requests, and Jira as part of day-to-day software development workflows.
- Built APIs and web systems deployed on AWS using React.js, TypeScript, Node.js, and Docker.

Education

Georgia Institute of Technology, Master of Science in Computer Science Jan 2024 – Dec 2025
GPA: 3.7/4.0 Specialization: Interactive Intelligence

Ahsanullah University of Science and Technology, BS in Computer Science and Engineering Nov 2015 – Feb 2020
GPA: 3.43/4.0

Publications and Research

KnitDraft / TMK-Knit | MAEKNIT Mar 2025 – Present

- Developing a human-in-the-loop AI system for industrial flat-knitting machine programming using Task–Method–Knowledge modeling to assist with initial machine-program generation while preserving expert validation.
- Designing a structured manufacturing knowledge pipeline using existing machine programs, CAD-derived specifications, measurement data, swatch testing, program revisions, manufacturing timing, and machine-specific knowledge.
- Exploring how symbolic manufacturing knowledge and LLM-based systems can augment specialized

manufacturing workers and reduce repetitive programming effort without removing expert control.

- Technologies: Python, LLMs, OpenAI APIs, TMK modeling, Odoo, industrial manufacturing data.

SAMI – Social Agent Mediated Interactions | Design Intelligence Lab Aug 2024 – Dec 2025

- Contributed to an AI social agent deployed in Georgia Tech's OMSCS program for ten semesters and used by more than 11,000 students.
- Co-designed a two-level metacognitive self-adaptation architecture integrating Knowledge-Based AI with LLMs to identify and correct reasoning errors.
- Integrated a Task–Method–Knowledge self-model enabling the agent to localize hallucination, omission, and misinterpretation errors.
- Built workflows combining Neo4j knowledge graphs, structured knowledge representations, and a solution library for automated adaptation.
- Technologies: Python, Neo4j, OpenAI LLMs, Docker, Knowledge-Based AI.

A Metacognitive Architecture for Correcting LLM Errors in AI Agents Jan 2026
IAAI-26: Emerging Applications of AI, AAAI 2026. Introduced a two-level metacognitive architecture using TMK-based self-modeling and an integrated Knowledge-Based AI and LLM approach to localize and correct hallucination and misinterpretation errors in deployed AI agents.

Correcting LLM Errors: A Metacognitive Architecture for ToM Adaptation in AI Agents Jan 2026
ToM4AI Workshop at AAAI 2026. Presented a metacognitive Theory-of-Mind adaptation framework evaluated on real-world student feedback, achieving a 75% successful adaptation rate across 20 cases.

Adaptable Social AI Agents May 2025
ToM4AI Workshop at AAAI 2025. Presented enhancements to the SAMI social AI agent using episodic self-explanation capabilities to support adaptive and context-dependent reasoning.

Hybrid Text Summarizer for Bangla Documents Nov 2020
International Journal of Computer Vision and Signal Processing, Vol. 10, No. 1 First Author
Developed a hybrid extractive summarization system for Bangla text using NLP and machine learning techniques.

Professional Service

- Reviewer, ACM Collective Intelligence Conference.
- Reviewer, Journal of Intelligent Manufacturing.
- Reviewer, IEEE INDICON 2026.
- Reviewer, 8th International Conference on Advances in Distributed Computing and Machine Learning (ICADCML 2027).